微积分 A1 第 9 次习题课参考答案　　定积分与变限积分

1. 计算下列积分：

(1) $\int_0^{\frac{\pi}{2}} \frac{\mathrm{d}x}{1+\sin^2 x}$　　(2) $\int_0^2 \lfloor e^x \rfloor \mathrm{d}x$ (其中·为下取整符号)

(3) $\int_0^{\pi} \sqrt{1-\sin x}\mathrm{d}x$　　(4) $\int_0^1 \left(x\int_{x^2}^1 e^{-t^2}\mathrm{d}t \right)\mathrm{d}x$

(5) $\int_{-\pi}^{\pi} \frac{\cos x}{1+e^x}\,\mathrm{d}x$　　(6) $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\mathrm{d}x$

(7) $\int_0^{\pi/2} \frac{1}{1+(\tan x)^{\sqrt{2}}}\mathrm{d}x$

解：(1) $\int_0^{\frac{\pi}{2}} \frac{\mathrm{d}x}{1+\sin^2 x} = \int_0^{\frac{\pi}{2}} \frac{1}{1+\sin^{-2} x}\cdot\frac{\mathrm{d}x}{\sin^2 x} = -\int_0^{\frac{\pi}{2}} \frac{\mathrm{d}\cot x}{2+\cot^2 x}$

$$= -\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{2}} \frac{\mathrm{d}\frac{\cot x}{\sqrt{2}}}{1+\left(\frac{\cot x}{\sqrt{2}}\right)^2} = -\frac{1}{\sqrt{2}}\arctan\left(\frac{\cot x}{\sqrt{2}}\right)\Bigg|_{x=0}^{\frac{\pi}{2}} = \frac{\sqrt{2}\pi}{4}.$$

(2) $\int_0^2 \lfloor e^x \rfloor \mathrm{d}x = \int_0^{\ln 2} 1\mathrm{d}x + \int_{\ln 2}^{\ln 3} 2\mathrm{d}x + \int_{\ln 3}^{\ln 4} 3\mathrm{d}x + \int_{\ln 4}^{\ln 5} 4\mathrm{d}x + \int_{\ln 5}^{\ln 6} 5\mathrm{d}x + \int_{\ln 6}^{\ln 7} 6\mathrm{d}x + \int_{\ln 7}^{2} 7\mathrm{d}x$

$$= \ln 2 + 2\ln\frac{3}{2} + 3\ln\frac{4}{3} + 4\ln\frac{5}{4} + 5\ln\frac{6}{5} + 6\ln\frac{7}{6} + 7(2-\ln 7) = 14 - \ln(7!)\text{。}$$

(3) $\int_0^{\pi} \sqrt{1-\sin x}\,\mathrm{d}x = \int_0^{\pi} \left|\sin\frac{x}{2} - \cos\frac{x}{2}\right|\mathrm{d}x$

$$= \int_0^{\frac{\pi}{2}} \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)\mathrm{d}x + \int_{\frac{\pi}{2}}^{\pi} \left(\sin\frac{x}{2} - \cos\frac{x}{2}\right)\mathrm{d}x = 4\sqrt{2} - 4.$$

(4) 令 $f(x) = \int_{x^2}^1 e^{-t^2}\mathrm{d}t$, 则 $f(1) = 0,\ f'(x) = -2xe^{-x^4}$, 于是

$$\int_0^1 \left(x\int_{x^2}^1 e^{-t^2}\mathrm{d}t \right)\mathrm{d}x = \int_0^1 xf(x)\,\mathrm{d}x = \frac{1}{2}\int_0^1 f(x)\,\mathrm{d}x^2 = \frac{1}{2}x^2 f(x)\Big|_0^1 - \frac{1}{2}\int_0^1 x^2 f'(x)\,\mathrm{d}x$$

$$= \int_0^1 x^3 e^{-x^4}\mathrm{d}x = -\frac{1}{4}e^{-x^4}\Big|_0^1 = \frac{1}{4}(1-e^{-1}).$$

(5) 令 $t = -x$, 则

$$I = \int_{-\pi}^{\pi} \frac{\cos x}{1+e^x}\mathrm{d}x = \int_{-\pi}^{\pi} \frac{\cos t}{1+e^{-t}}\,\mathrm{d}t = \int_{-\pi}^{\pi} \frac{e^t\cos t}{1+e^t}\,\mathrm{d}t,$$

$$I = \frac{1}{2}\left(\int_{-\pi}^{\pi} \frac{\cos x}{1+e^x}\,\mathrm{d}x + \int_{-\pi}^{\pi} \frac{e^x\cos x}{1+e^x}\,\mathrm{d}x \right) = \frac{1}{2}\int_{-\pi}^{\pi} \cos x\,\mathrm{d}x = 0.$$

(6) **解法一**：令 $t=\pi-x$, 则

$$I=\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}\mathrm{d}x=\int_0^{\pi}\frac{(\pi-t)\sin t}{1+\cos^2 t}\mathrm{d}t=\pi\int_0^{\pi}\frac{\sin t}{1+\cos^2 t}\mathrm{d}t-I,$$

$$I=\frac{\pi}{2}\int_0^{\pi}\frac{\sin x}{1+\cos^2 x}\mathrm{d}x=-\frac{\pi}{2}\int_0^{\pi}\frac{\mathrm{d}\cos x}{1+\cos^2 x}=-\frac{\pi}{2}\arctan(\cos x)\Big|_0^{\pi}=\frac{\pi^2}{4}.$$

注：设 $f,g\in C[0,a]$. 若$\exists l$, 使得在 $[0,a]$ 上有$f(x)=f(a-x), g(x)+g(a-x)=l$, 则可证明

$$\int_0^a f(x)g(x)\mathrm{d}x=\frac{l}{2}\int_0^a f(x)\mathrm{d}x.$$

解法二：$\displaystyle\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}\mathrm{d}x=-\int_0^{\pi}x\,\mathrm{d}\arctan(\cos x)$

$$=-x\arctan(\cos x)\Big|_0^{\pi}+\int_0^{\pi}\arctan(\cos x)\mathrm{d}x=\frac{\pi^2}{4}+\int_0^{\pi}\arctan(\cos x)\mathrm{d}x,$$

令 $t=x-\dfrac{\pi}{2}$, 则

$$\int_0^{\pi}\arctan(\cos x)\mathrm{d}x=\int_{-\pi/2}^{\pi/2}\arctan(\sin t)\mathrm{d}t=0.\text{（奇函数）}$$

因此 $\displaystyle\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}\mathrm{d}x=\frac{\pi^2}{4}.$

(7) $f(x)=\dfrac{1}{1+(\tan x)^{\sqrt{2}}}$, 令 $f(\dfrac{\pi}{2})=\lim\limits_{x\to\frac{\pi}{2}^-}f(\dfrac{\pi}{2})=0$, 则 $f\in C[0,\dfrac{\pi}{2}]\subset R[0,\dfrac{\pi}{2}]$. 令 $t=\dfrac{\pi}{2}-x$, 得

$$\int_0^{\pi/2}\frac{1}{1+(\tan x)^{\sqrt{2}}}\mathrm{d}x=\int_0^{\pi/2}\frac{1}{1+(\cot t)^{\sqrt{2}}}\mathrm{d}t=\int_0^{\pi/2}\frac{(\tan t)^{\sqrt{2}}}{1+(\tan t)^{\sqrt{2}}}\mathrm{d}t.$$

因此 $\displaystyle\int_0^{\pi/2}\frac{1}{1+(\tan x)^{\sqrt{2}}}\mathrm{d}x=\frac{1}{2}\left(\int_0^{\pi/2}\frac{1}{1+(\tan x)^{\sqrt{2}}}\mathrm{d}x+\int_0^{\pi/2}\frac{(\tan t)^{\sqrt{2}}}{1+(\tan t)^{\sqrt{2}}}\mathrm{d}t\right)=\frac{1}{2}\int_0^{\pi/2}\mathrm{d}x=\frac{\pi}{4}.$

2. 已知 $f(x)=\displaystyle\int_0^x f(t)\mathrm{d}t, \forall x\in\mathbb{R}$. 求 $f(x)$.

解：等式 $f(x)=\displaystyle\int_0^x f(t)\mathrm{d}t$ 两边对 x 求导,得 $f'(x)=f(x)$. 于是

$$\left(e^{-x}f(x)\right)'=e^{-x}(f'(x)-f(x))\equiv 0,\ e^{-x}f(x)=c,\ f(x)=ce^x.$$

而 $f(0)=\displaystyle\int_0^0 f(t)\mathrm{d}t=0$, 所以 $c=0,\ f(x)\equiv 0.$

3. 已知 $f(x)\in C^1(-1,1)$, 求 $\lim\limits_{x\to 0^+}\dfrac{1}{x^4}\int_{-x}^{x}[f(t^2+x)-f(t^2+\sin x)]\mathrm{d}t$.

解： 由积分第一中值定理，$\exists \xi_x\in[-x,x], s.t.$

$$\int_{-x}^{x}[f(t^2+x)-f(t^2+\sin x)]\mathrm{d}t=2x[f(\xi_x^2+x)-f(\xi_x^2+\sin x)].$$

由 Lagrange 中值定理，$\exists \eta_x\in(\xi_x^2+\sin x,\xi_x^2+x), s.t.$

$$f(\xi_x^2+x)-f(\xi_x^2+\sin x)=f'(\eta_x)(x-\sin x).$$

注意到 $\lim\limits_{x\to 0^+}\xi_x=0, \lim\limits_{x\to 0^+}\eta_x=0$, 而 $f(x)\in C^1(-1,1)$, 因此

$$\lim_{x\to 0^+}\frac{1}{x^4}\int_{-x}^{x}[f(t^2+x)-f(t^2-\sin x)]\mathrm{d}t=\lim_{x\to 0^+}\frac{2xf'(\eta_x)(x-\sin x)}{x^4}$$

$$=2\lim_{x\to 0^+}f'(\eta_x)\cdot\lim_{x\to 0^+}\frac{x-\sin x}{x^3}=2f'(0)\cdot\frac{1}{6}=\frac{1}{3}f'(0).\square$$

4. 已知 $f(x)$ 连续且 $\lim\limits_{x\to 0}\dfrac{f(x)}{x}=a$, 定义 $\varphi(x)=\int_0^1 f(x^2t)\mathrm{d}t$, 求 $\varphi'(x)$，并讨论 $\varphi'(x)$ 在点 $x=0$ 的连续性。

解： $f(x)$ 连续,则 $f(0)=\lim\limits_{x\to 0}f(x)=\lim\limits_{x\to 0}x\cdot\lim\limits_{x\to 0}\dfrac{f(x)}{x}=0\cdot a=0.$ 于是

$$\varphi(0)=\int_0^1 f(0)\mathrm{d}t=0.$$

当 $x\neq 0$ 时，令 $s=x^2t$, 则

$$\varphi(x)=\int_0^1 f(x^2t)\mathrm{d}t=\frac{1}{x^2}\int_0^{x^2}f(s)\mathrm{d}s.$$

已知 $\lim\limits_{x\to 0}\dfrac{f(x)}{x}=a$, 于是

$$\varphi'(0)=\lim_{x\to 0}\frac{\varphi(x)-\varphi(0)}{x-0}=\lim_{x\to 0}\frac{\int_0^{x^2}f(s)\mathrm{d}s}{x^3}=\lim_{x\to 0}\frac{2xf(x^2)}{3x^2}=\frac{2}{3}\lim_{x\to 0}x\cdot\lim_{x\to 0}\frac{f(x^2)}{x^2}=0.$$

$$\varphi'(x)=\frac{x^2\cdot f(x^2)\cdot 2x-2x\int_0^{x^2}f(s)\mathrm{d}s}{x^4}=\frac{2f(x^2)}{x}-\frac{2\int_0^{x^2}f(s)\mathrm{d}s}{x^3},\ x\neq 0.$$

而 $\lim\limits_{x\to 0}\varphi'(x)=2\lim\limits_{x\to 0}x\cdot\lim\limits_{x\to 0}\dfrac{f(x^2)}{x^2}-2\lim\limits_{x\to 0}\dfrac{\int_0^{x^2}f(s)\mathrm{d}s}{x^3}=0-\lim\limits_{x\to 0}\dfrac{2xf(x^2)}{3x^2}=0,$

故 $\varphi'(x)$ 在点 $x=0$ 连续。$\square$

5. $f\in C[0,+\infty)\ f\in C^1(0,+\infty),\int_0^x tf(t)\mathrm{d}t=ax\int_0^x f(t)\mathrm{d}t,\forall x\geq 0.$ 求 $f(x).$

解： $\int_0^x tf(t)\mathrm{d}t=ax\int_0^x f(t)\mathrm{d}t$ 两边对 x 求导，整理得

$$(1-a)xf(x)=a\int_0^x f(t)\mathrm{d}t,\qquad \forall x>0.$$

两边再对 x 求导，得

$$(1-a)xf'(x)=(2a-1)f(x),\qquad \forall x>0.$$

若 $a=1,$ 则 $f(x)\equiv 0.$

若 $a\neq 1,$ 令 $F(x)=x^{(2a-1)/(a-1)}f(x),$ 则

$$F'(x)=x^{(2a-1)/(a-1)}\left(f'(x)-\frac{2a-1}{(1-a)x}f(x)\right)=0,\qquad \forall x>0.$$

因此 $F(x)\equiv C,\ f(x)=Cx^{(2a-1)/(1-a)},C\in\mathbb{R}.$

6. 试求 $\lim\limits_{n\to+\infty}\dfrac{1}{\ln n}\int_0^{\pi/2}\dfrac{\sin^2 nx}{\sin x}\mathrm{d}x.$

解： $\sin^2(n+1)x-\sin^2 nx=\big[\sin(n+1)x-\sin nx\big]\big[\sin(n+1)x+\sin nx\big]$

$$=2\sin\frac{x}{2}\cos\frac{(2n+1)x}{2}\cdot 2\sin\frac{(2n+1)x}{2}\cos\frac{x}{2}=\sin x\sin(2n+1)x.$$

于是, 由 Stolz 公式得

$$\lim_{n\to+\infty}\frac{1}{\ln n}\int_0^{\pi/2}\frac{\sin^2 nx}{\sin x}\mathrm{d}x=\lim_{n\to+\infty}\frac{\int_0^{\pi/2}\sin(2n+1)x\,\mathrm{d}x}{\ln(n+1)-\ln n}=\lim_{n\to+\infty}\frac{\frac{1}{2n+1}}{\ln(1+\frac{1}{n})}=\frac{1}{2}.\ \square$$

7. 证明：$\forall x\geq 0,\int_{\frac{-\pi}{2}}^x\dfrac{\cos t}{1+e^t}\mathrm{d}t\geq 0.$

证明： 对任意非负整数 k 及 $s\in[2k\pi-\frac{\pi}{2},2k\pi+\frac{\pi}{2}],$ 有 $0\leq\dfrac{\cos s}{1+e^{s+\pi}}\leq\dfrac{\cos s}{1+e^s}.$ 令 $t=s+\pi,$ 有

$$\int_{2k\pi+\frac{\pi}{2}}^{2k\pi+\frac{3\pi}{2}}\frac{\cos t}{1+e^t}\mathrm{d}t=\int_{2k\pi-\frac{\pi}{2}}^{2k\pi+\frac{\pi}{2}}\frac{-\cos s}{1+e^{s+\pi}}\mathrm{d}s\geq-\int_{2k\pi-\frac{\pi}{2}}^{2k\pi+\frac{\pi}{2}}\frac{\cos s}{1+e^s}\mathrm{d}s=-\int_{2k\pi-\frac{\pi}{2}}^{2k\pi+\frac{\pi}{2}}\frac{\cos t}{1+e^t}\mathrm{d}t.$$

因此对任意非负整数 $k,$ 有

$$\int_{2k\pi-\frac{\pi}{2}}^{2k\pi+\frac{3\pi}{2}}\frac{\cos t}{1+e^t}\mathrm{d}t=\int_{2k\pi-\frac{\pi}{2}}^{2k\pi+\frac{\pi}{2}}\frac{\cos t}{1+e^t}\mathrm{d}t+\int_{2k\pi+\frac{\pi}{2}}^{2k\pi+\frac{3\pi}{2}}\frac{\cos t}{1+e^t}\mathrm{d}t\geq 0.$$

进一步有

$$\int_{-\frac{\pi}{2}}^{2k\pi+\frac{3\pi}{2}}\frac{\cos t}{1+e^t}\mathrm{d}t\geq 0,\quad \forall k\geq 0.$$

$\forall x\geq-\frac{\pi}{2}$, 存在非负整数 k, 使得 $2k\pi-\frac{\pi}{2}\leq x<2k\pi+\frac{3\pi}{2}$. 若 $2k\pi-\frac{\pi}{2}\leq x<2k\pi+\frac{\pi}{2}$,

则

$$\int_{-\frac{\pi}{2}}^{x}\frac{\cos t}{1+e^t}\mathrm{d}t=\int_{-\frac{\pi}{2}}^{2k\pi-\frac{\pi}{2}}\frac{\cos t}{1+e^t}\mathrm{d}t+\int_{2k\pi-\frac{\pi}{2}}^{x}\frac{\cos t}{1+e^t}\mathrm{d}t\geq 0.$$

若 $2k\pi+\frac{\pi}{2}\leq x<2k\pi+\frac{3\pi}{2}$, 则

$$\int_{-\frac{\pi}{2}}^{x}\frac{\cos t}{1+e^t}\mathrm{d}t\geq\int_{-\frac{\pi}{2}}^{x}\frac{\cos t}{1+e^t}\mathrm{d}t+\int_{x}^{2k\pi+\frac{3\pi}{2}}\frac{\cos t}{1+e^t}\mathrm{d}t=\int_{-\frac{\pi}{2}}^{2k\pi+\frac{3\pi}{2}}\frac{\cos t}{1+e^t}\mathrm{d}t\geq 0.\square$$

8. $f\in C^1[0,a],\ f(0)=0.$ 证明：$\int_0^a|f(x)f'(x)|\mathrm{d}x\leq\frac{a}{2}\int_0^a\left(f'(x)\right)^2\mathrm{d}x.$

证明：令 $g(x)=\int_0^x|f'(t)|\mathrm{d}t, x\in[0,a]$, 则 $g(0)=0,\ g'(x)=|f'(x)|$. 又由 $f(0)=0$ 得，

$$|f(x)|=\left|f(0)+\int_0^x f'(t)\mathrm{d}t\right|\leq\int_0^x|f'(t)|\mathrm{d}t=g(x),\quad\forall x\in[0,a].$$

于是，

$$\int_0^a|f(x)f'(x)|\mathrm{d}x\leq\int_0^a g(x)g'(x)\mathrm{d}x=\frac{1}{2}g^2(x)\Big|_{x=0}^{a}=\frac{1}{2}g^2(a)$$

$$=\frac{1}{2}\left(\int_0^a|f'(t)|\mathrm{d}t\right)^2\leq\frac{1}{2}\int_0^a|f'(t)|^2\mathrm{d}t\cdot\int_0^a 1^2\mathrm{d}t=\frac{a}{2}\int_0^a|f'(t)|^2\mathrm{d}t.\square$$

9. $f\in C^1[0,1]$, 证明：

（1）$|f(x)|\leq\int_0^1|f(t)|\mathrm{d}t+\int_0^1|f'(t)|\mathrm{d}t,\forall x\in[0,1];$

（2）$\left|f(\frac{1}{2})\right|\leq\int_0^1|f(t)|\mathrm{d}t+\frac{1}{2}\int_0^1|f'(t)|\mathrm{d}t.$

证明：（1）$\forall x\in[0,1]$, 由积分第一中值定理，$\exists\xi\in[0,x], s.t.$

$$f(x)-\int_0^1 f(t)\mathrm{d}t = f(x)-f(\xi)=\int_\xi^1 f'(t)\mathrm{d}t$$

于是

$$\begin{aligned}|f(x)| &\le \left|\int_0^1 f(t)\mathrm{d}t\right|+\left|\int_\xi^1 f'(t)\mathrm{d}t\right| \\ &\le \int_0^1 |f(t)|\mathrm{d}t+\int_\xi^1 |f'(t)|\mathrm{d}t \le \int_0^1 |f(t)|\mathrm{d}t+\int_0^1 |f'(t)|\mathrm{d}t.\end{aligned}$$

（2）由积分第一中值定理，$\exists \xi_1\in[0,\frac{1}{2}],\xi_2\in[\frac{1}{2},1], s.t.$

$$\frac{1}{2}f(\frac{1}{2})-\int_0^{1/2} f(t)\mathrm{d}t=\frac{1}{2}f(\frac{1}{2})-\frac{1}{2}f(\xi_1)=\frac{1}{2}\int_{\xi_1}^{1/2} f'(t)\mathrm{d}t,$$

$$\frac{1}{2}f(\frac{1}{2})-\int_{1/2}^{1} f(t)\mathrm{d}t=\frac{1}{2}f(\frac{1}{2})-\frac{1}{2}f(\xi_2)=-\frac{1}{2}\int_{1/2}^{\xi_2} f'(t)\mathrm{d}t.$$

两式相加，得

$$f(\frac{1}{2})-\int_0^1 f(t)\mathrm{d}t=\frac{1}{2}\int_{\xi_1}^{1/2} f'(t)\mathrm{d}t-\frac{1}{2}\int_{1/2}^{\xi_2} f'(t)\mathrm{d}t.$$

因此，

$$\begin{aligned}\left|f(\frac{1}{2})\right| &\le \left|\int_0^1 f(t)\mathrm{d}t\right|+\frac{1}{2}\left|\int_{\xi_1}^{1/2} f'(t)\mathrm{d}t\right|+\frac{1}{2}\left|\int_{1/2}^{\xi_2} f'(t)\mathrm{d}t\right| \\ &\le \int_0^1 |f(t)|\mathrm{d}t+\frac{1}{2}\int_{\xi_1}^{1/2}|f'(t)|\mathrm{d}t+\frac{1}{2}\int_{1/2}^{\xi_2}|f'(t)|\mathrm{d}t \\ &\le \int_0^1 |f(t)|\mathrm{d}t+\frac{1}{2}\int_0^{1/2}|f'(t)|\mathrm{d}t+\frac{1}{2}\int_{1/2}^{1}|f'(t)|\mathrm{d}t \\ &= \int_0^1 |f(t)|\mathrm{d}t+\frac{1}{2}\int_0^1|f'(t)|\mathrm{d}t.\end{aligned}$$

10. 证明：$\left(\int_0^1 xe^{x^2}\cos x\mathrm{d}x\right)^2+\left(\int_0^1 xe^{x^2}\sin x\mathrm{d}x\right)^2\le\frac{1}{4}(e-1)^2.$

证明：由积分估值不等式和 Cauchy 不等式，有

$$\begin{aligned}\left(\int_0^1 xe^{x^2}\cos x\mathrm{d}x\right)^2 &= \left|\int_0^1 xe^{x^2}\cos x\mathrm{d}x\right|^2 \le \left(\int_0^1 xe^{x^2}|\cos x|\mathrm{d}x\right)^2 \\ &= \left(\int_0^1 \sqrt{xe^{x^2}}\cdot\sqrt{xe^{x^2}}\,|\cos x|\mathrm{d}x\right)^2 \le \int_0^1 xe^{x^2}\mathrm{d}x\cdot\int_0^1 xe^{x^2}\cos^2 x\mathrm{d}x.\end{aligned}$$

同理，有

$$\left(\int_0^1 xe^{x^2}\sin x\mathrm{d}x\right)^2\le\int_0^1 xe^{x^2}\mathrm{d}x\cdot\int_0^1 xe^{x^2}\sin^2 x\mathrm{d}x.$$

两式相加，得

$$\left(\int_0^1 xe^{x^2}\cos x\mathrm{d}x\right)^2+\left(\int_0^1 xe^{x^2}\sin x\mathrm{d}x\right)^2\le\left(\int_0^1 xe^{x^2}\mathrm{d}x\right)^2=\left(\frac{1}{2}e^{x^2}\Big|_0^1\right)^2=\frac{1}{4}(e-1)^2.\square$$